THE LEVEL SETS OF THE COUNT OF GEOMETRICALLY IRREDUCIBLE FIBRE COMPONENTS ARE LOCALLY CONSTRUCTIBLE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 7793-7804 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 7806-7829, the complete author proof (22 body lines).

Editorial note. Navigation stubs: 4 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Geometrically irreducible components of a fibre, constructibility of finite-presentation conditions, and base change to an algebraic closure; the encoding is retained.

Source reference. Original lemma lemma-base-change-fibres-nr-geometrically-irreducible-components: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7453–7466. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-nr-geom-irreducible-components-good: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7741–7749. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma limits-lemma-descend-finite-presentation: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7829–7829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma topology-lemma-characterize-constructible-Noetherian: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 7829–7829. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : X \to Y$ be a morphism of schemes. Let $n_{X/Y}$ be the function on Y counting the numbers of geometrically irreducible components of fibres of f introduced in Lemma lemma-base-change-fibres-nr-geometrically-irreducible-components. Assume f of finite presentation. Then the level sets*

$$E_n = \{y \in Y \mid n_{X/Y}(y) = n\}$$

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of $n_{X/Y}$ are locally constructible in Y.

Proof. Fix n. Let $y \in Y$. We have to show that there exists an open neighbourhood V of y in Y such that $E_n \cap V$ is constructible in V. Thus we may assume that Y is affine. Write $Y = \mathrm{Spec}(A)$ and $A = \mathrm{colim}\, A_i$ as a directed limit of finite type $\mathbf{Z}$-algebras. By Limits, Lemma limits-lemma-descend-finite-presentation we can find an i and a morphism $f_i : X_i \to \mathrm{Spec}(A_i)$ of finite presentation whose base change to Y recovers f. By Lemma lemma-base-change-fibres-nr-geometrically-irreducible-components it suffices to prove the lemma for f_i. Thus we reduce to the case where Y is the spectrum of a Noetherian ring.

We will use the criterion of Topology, Lemma topology-lemma-characterize-constructible-Noetherian to prove that E_n is constructible in case Y is a Noetherian scheme. To see this let $Z \subset Y$ be an irreducible closed subscheme. We have to show that $E_n \cap Z$ either contains a nonempty open subset or is not dense in Z. Let $\xi \in Z$ be the generic point. Then Lemma lemma-nr-geom-irreducible-components-good shows that $n_{X/Y}$ is constant in a neighbourhood of ξ in Z. This clearly implies what we want. $\square$